

ORIGINAL RESEARCH

Coronary Microvascular Resistance as a Predictor of the Placebo-Controlled Response to Percutaneous Coronary Intervention

Microvascular-Stratified Analysis of ORBITA

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ABSTRACT

BACKGROUND The first placebo-controlled trial of percutaneous coronary intervention (PCI), ORBITA (Objective Randomised Blinded Investigation With Optimal Medical Therapy of Angioplasty in Stable Angina; [NCT02062593](#)), showed minimal symptom benefit with PCI. No placebo-controlled data describing the relationship between microvascular resistance (MVR) and PCI exist. The authors hypothesized that patients with low MVR would derive the greatest benefit from PCI.

OBJECTIVES The aims of this study were to compute MVR in patients recruited for ORBITA and to evaluate interactions with prespecified endpoints.

METHODS Hyperemic MVR was calculated using computational fluid dynamics (CFD) and compared against placebo-controlled changes in treadmill exercise time, patient-reported symptoms, physician-assessed symptoms, and dobutamine stress echocardiography scores at 6-week follow-up.

RESULTS MVR_{CFD} was computed for 131 patients (66 undergoing PCI and 65 placebo). Median MVR_{CFD} was 1.38 mm Hg · min/mL (Q1-Q3: 0.89-2.09 mm Hg · min/mL). Baseline exercise time correlated with MVR_{CFD} (ordinal correlation coefficient = 0.20; 95% credible interval [CrI]: 0.18-0.22). For patients with low (20th centile) MVR_{CFD} , PCI increased exercise time by 48 seconds vs placebo (95% CrI: 6-92 seconds; Pr = 98.5%). Exercise time did not improve for patients with high (80th centile) MVR_{CFD} (16 seconds; 95% CrI: -29 to 61 seconds; probability of significant difference [Pr] = 75.2%), but evidence for an interaction was modest ($Pr_{interaction}$ = 83.1%). Low MVR_{CFD} was also associated with a placebo-controlled benefit of PCI for likelihood of complete freedom from angina (Pr = 98.8%) and improvements in angina frequency (Pr = 97.8%) and stress echocardiography scores (Pr = 99.9%).

CONCLUSIONS The placebo-controlled benefit of PCI was greater in patients with lower MVR_{CFD} , but interactions with symptom-based endpoints were modest. For patients with severe single-vessel disease taking optimal medical therapy, microvascular dysfunction may attenuate the functional and symptomatic benefits of PCI.

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ABBREVIATIONS
AND ACRONYMS

CAD = coronary artery disease

CCS = Canadian Cardiovascular Society

CFD = computational fluid dynamics

CMD = coronary microvascular dysfunction

CrI = credible interval

DSE = dobutamine stress echocardiography

FFR = fractional flow reserve

iFR = instantaneous wave-free ratio

MVR = microvascular resistance

PCI = percutaneous coronary intervention

Pr = probability of significant benefit

SAQ = Seattle Angina Questionnaire

Percutaneous coronary intervention (PCI) is a mainstay of treatment for anginal symptoms caused by flow-limiting epicardial coronary artery disease (CAD),¹ and this practice is supported by several previous unblinded trials.²⁻⁴ The first placebo-controlled trial of PCI, ORBITA (Objective Randomised Blinded Investigation With Optimal Medical Therapy of Angioplasty in Stable Angina), challenged this belief. ORBITA recruited patients with severe single-vessel epicardial disease already treated with maximal antianginal medication and showed that PCI did not lead to significant improvements in exercise capacity or reductions in symptoms compared with a placebo intervention.⁵ This was despite the inclusion of physiologically significant lesions (mean fractional flow reserve [FFR] 0.69) and objective evidence confirming myocardial ischemia in almost all recruited patients. The lack of a significant benefit of PCI was independent of physiological lesion

severity⁶ and disease pattern,⁷ with no evidence of an interaction between FFR or instantaneous wave-free ratio (iFR) and the placebo-controlled efficacy of PCI with exercise time and both patient-reported and physician-assessed symptoms. PCI in ORBITA did, however, improve dobutamine stress echocardiography (DSE) score. A subsequent subanalysis showed that this effect was predicted by invasive physiology, with greater improvements in DSE score with decreasing FFR and iFR.⁶

The somewhat unexpected findings were suggested to have resulted from a complex pathway in the progression from luminal stenosis to flow-limiting epicardial disease, stress myocardial ischemia, and, ultimately, patient symptoms.⁸ Limitations in diagnostic techniques may also have contributed; the prerandomization ischemia testing did not include a focused assessment of coronary microvascular dysfunction (CMD). This may have been important given the prevalence of CMD across

the spectrum of chronic coronary syndromes^{1,9} and remains a limitation of the original ORBITA study⁵ and subsequent subanalyses.^{8,10} Established measures for quantifying microvascular resistance (MVR) include Doppler-derived hyperemic MVR and bolus thermodilution-derived index of MVR. However, these are indirect surrogates of flow and, even in expert hands, correlate poorly with each other.¹¹ Invasive quantification of absolute hyperemic MVR is the emerging gold-standard assessment but requires a dedicated continuous infusion thermodilution catheter,¹² which was not used in the ORBITA trial. Recent developments in computational fluid dynamics (CFD) have allowed the post hoc computation of absolute coronary blood flow and MVR (MVR_{CFD}) from standard angiography and pressure wire data.¹³ In this retrospective analysis of the ORBITA data set, we computed MVR_{CFD} and assessed its impact on the placebo-controlled effect of PCI on the primary and secondary endpoints of ORBITA.

METHODS

ORBITA DESIGN AND CLINICAL DATA COLLECTION.

ORBITA was a double-blind, randomized, controlled trial of PCI vs a placebo procedure in patients aged 18 to 85 years with stable angina and angiographically severe single-vessel coronary disease ($\geq 70\%$ stenosis). Patients were recruited across 5 study sites in the United Kingdom and randomized between January 6, 2014, and August 11, 2017. The full design has been previously reported.⁵ In brief, enrolled patients entered an initial 6-week medical optimization phase, during which antianginal medications were up-titrated followed by prerandomization assessment including: 1) cardiopulmonary exercise testing with smoothed modified Bruce protocol¹⁴; 2) patient-reported symptoms using the Seattle Angina Questionnaire (SAQ); 3) patient-reported symptoms using the EQ-5D-5L questionnaire; 4) physician-assessed grading of angina severity (Canadian Cardiovascular Society [CCS] class); and 5) DSE, with each scan examined twice by 6 blinded imaging cardiology

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consultants. Wall motion was scored to represent the number of hypokinetic segments.⁶

Enrolled patients then underwent blinded invasive coronary angiography, including blinded FFR and iFR assessment performed twice, according to standard techniques.¹⁵ After invasive physiological assessment, patients were randomized 1:1 to either PCI or a placebo procedure. For patients allocated to intervention, the use of intravascular imaging was encouraged, and post-PCI physiology was measured as part of the study protocol with a mandate to achieve complete revascularization. Patients then entered a blinded 6-week follow-up period, at the end of which all prerandomization assessments were repeated. The study was approved by a national ethics committee and was compliant with the Declaration of Helsinki, and all patients provided informed written consent. The ORBITA trial is registered at ClinicalTrials.gov (NCT02062593).

VIRTUAL PHYSIOLOGICAL ASSESSMENT. Offline virtual physiological quantification of MVR_{CFD} was performed at the University of Sheffield, using virtuQ, a CFD software package developed in house. This technique reconstructs a virtual 3-dimensional coronary artery, representative of patient-specific coronary anatomy, and applies invasively measured pressures to derive MVR_{CFD} (Central Illustration). A complete description of the virtuQ methodology, along with benchtop and in vivo validation, has been reported previously.¹³ Two angiographic projections of the vessel of interest, acquired at least 30° apart and during end-diastole, were manually selected and coregistered. The vessel centerline and luminal border were then traced semiautomatically in both images, with manual correction if required. The reconstructed vessel inlet and outlet corresponded to the location of proximal and distal pressure measurements during invasive FFR assessment. All reconstructions were performed by an operator blinded to invasive physiology and then checked for anatomical accuracy by a blinded interventional cardiologist with experience in computer modeling of coronary arteries.

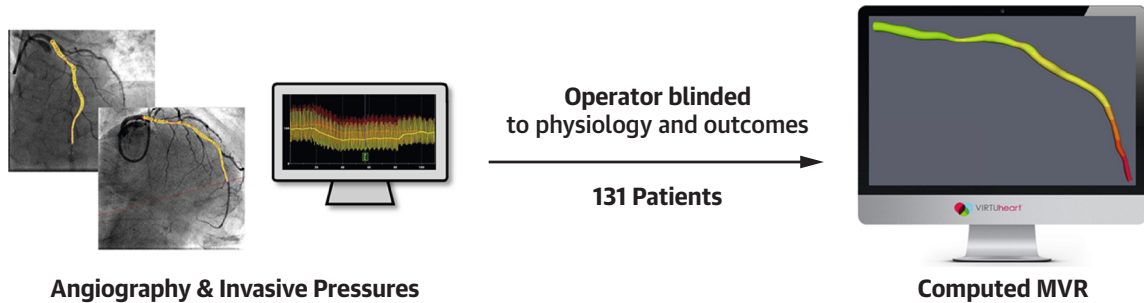
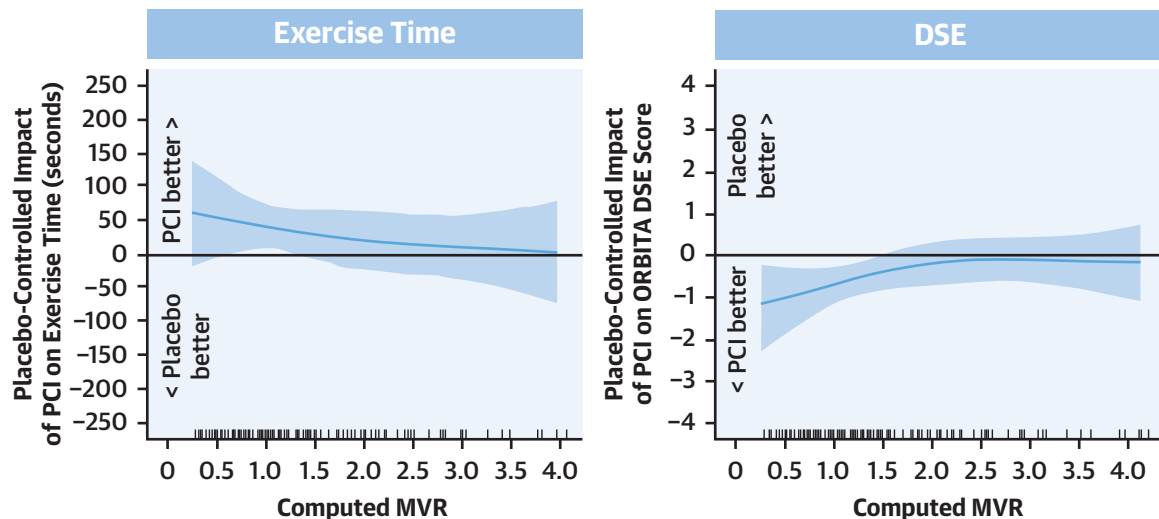
Raw invasive physiological data from each reconstructed case were interrogated using custom proprietary software (University of Sheffield) to compile transient pressure signals across several cardiac cycles. Transient pressure signals under hyperemic conditions were compiled across a target of 10 and minimum of 5 continuous cardiac cycles. This was performed for both FFR measurements and then averaged. The invasive pressure data defined inlet and outlet boundary conditions for CFD analysis.

Simulations were performed using standard blood rheological parameters (density 1,056 kg/m³, viscosity 0.0035 Pa · s) and simulated regionalized side branch flow according to vessel taper, with a flow-diameter scaling exponent of 2.33.^{16,17} Absolute MVR_{CFD} (mm Hg · min/mL) was calculated by dividing distal pressure by distal flow. Thus, MVR_{CFD} was computed in the distal vessel, corresponding to the location of the pressure wire transducer, and is specific for microvascular physiology even in the context of epicardial stenoses. This contrasts with continuous thermodilution,¹² which considers proximal resistance (corresponding to a higher subtended myocardial mass) and gives lower resistance values that are not microvascular specific in the context of epicardial stenoses. All research team members involved in processing data for the purpose of CFD simulations did so fully blinded to the clinical outcomes.

STATISTICAL ANALYSIS. This MVR-stratified sub-analysis of ORBITA considered patients with invasive physiological data taken at prerandomization under hyperemic conditions. Summary statistics are presented as frequency (percentage), mean ± SD, or median (Q1-Q3) as appropriate. Correlation between prerandomization assessments with MVR_{CFD} was assessed using Somer's D (D_{xy}).

To assess the impact of MVR_{CFD} on the placebo-controlled response to PCI, ordinal (proportional odds) regression modeling was used. The follow-up value was conditioned on the prerandomization value, with nonlinearity allowed using a restricted cubic spline with 3 knots (at the 10th, 50th, and 90th centiles), and randomization arm and MVR_{CFD} . The randomization arm was allowed to interact with MVR_{CFD} . For freedom from angina, a logistic regression model was used. All regression modeling was performed within a Bayesian framework. For the intercepts, priors were induced by a Dirichlet distribution on the cell probabilities, where all covariates were set to their means. This enforced a strict ordering of the intercepts, as they are defined by logits of cell probabilities accumulated over increasing values of the response. For the treatment effect (log OR), the prior was normal with mean 0 and SD chosen such that the prior probability of OR <0.25 equaled the prior probability of OR >4, with both equaling 0.05. Thus, the analysis was skeptical about the treatment effect being large in either direction. For covariates, a virtually flat prior was used (ie, a distribution with mean 0 and SD 100 on a normalized covariate scale).

For all endpoints, the Markov-chain Monte Carlo process used 4 chains with 20,000 samples (with

CENTRAL ILLUSTRATION Study Summary and Main Findings**Relationship Between Microvascular Resistance (MVR) and Benefit of PCI****A Microvascular Resistance Computed in ORBITA Cohort****B Comparison With Clinical Outcomes**

Low computed MVR also associated with benefit of PCI for:

- Angina freedom (Probability = 98.8%)
- Angina frequency (Probability = 97.8%)

- High MVR attenuated symptomatic and functional benefits of PCI.
- Microvascular disease significant despite optimal medical therapy.
- Consideration of microvascular disease may be useful before PCI.

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(A) Microvascular resistance (MVR) computed in ORBITA (Objective Randomised Blinded Investigation With Optimal Medical Therapy of Angioplasty in Stable Angina) cohort, summarizing the clinical data required to compute MVR, the number of included patients, and the simulation protocol. (B) Comparison with clinical outcomes, showing regression plots demonstrating the relationship between change in prerandomization to follow-up total exercise time and dobutamine stress echocardiography (DSE) score against prerandomization MVR by randomization arm. At progressively lower MVR values, there was a progressively larger difference between percutaneous coronary intervention (PCI) and placebo on the endpoint of exercise time (probability of interaction = 83.1%) and DSE score improvement (probability of interaction = 96.5%).

10,000 burn-in iterations). We calculated the relative explained variance of the interaction between the randomization arm and MVR_{CFD} along with its probability value ($Pr_{interaction}$). To aid clinical interpretability, we also present a contrast derived from the model of a “typical” patient at the 20th against the 80th centile of MVR_{CFD} . This may be interpreted as the expected placebo-controlled benefit of PCI on outcome variables for patients with low and high MVR_{CFD} , respectively. Results are therefore reported as the median placebo-controlled effect of PCI, with associated 95% credible interval (CrI) and probability of significant benefit favoring PCI vs placebo (Pr) for low- and high- MVR_{CFD} patients, followed by quantification of the interactional effect ($Pr_{interaction}$) across the entire patient cohort. The regression model specifications, concentration parameters for Dirichlet priors, output, chain mixing plots, and density plots are included in the [Supplemental Appendix](#). All analyses were performed in R version 4.4.1 (R Foundation for Statistical Computing), using the rmsb package for Bayesian modeling and ggplot2 for graphs.

RESULTS

Among the 200 patients in ORBITA, 131 (66 undergoing PCI, 65 undergoing placebo intervention) were included in this virtual physiology subanalysis. Of the excluded patients, 39 did not have sufficient angiographic data, 18 had coronary anatomy unsuitable for vessel reconstruction, 7 lacked hyperemic pressure data, and 5 successfully reconstructed vessels failed to reach CFD simulation convergence ([Supplemental Figure 1](#)).

PATIENT DEMOGRAPHICS. The baseline data on the patients included in this subanalysis are shown in [Table 1](#). More than 95% of patients had physician-assessed CCS class II ($n = 82$ of 131 [62.6%]) or III ($n = 46$ of 131 [35.1%]) angina severity at enrollment.

MEDICAL THERAPY. At prerandomization, 98.5% of patients (129 of 131) were taking dual antiplatelet therapy, with 2 patients intolerant to aspirin. Of the 131 included patients, 120 (91.6%) were taking calcium-channel antagonists, while 105 (80.2%) were taking beta-blockers. After the medical optimization phase, 28 of the 131 patients (21.4%) had CCS class 0 or I angina severity ([Supplemental Tables 1 to 3](#)).

PROCEDURAL DEMOGRAPHICS. Of the 131 included patients, almost all ($n = 128$ [97.7%]) had at least 1 positive ischemia test result at the time of randomization, which included a pre-enrollment investigation, research DSE, or physiologically significant

TABLE 1 Patient Demographics at Enrollment

	Percutaneous Coronary Intervention (n = 66)	Placebo (n = 65)	Complete Group (N = 131)
Age, y	64.7 ± 9.3	67.9 ± 7.6	66.3 ± 8.6
Male	44 (66.7)	47 (72.3)	91 (69.5)
Body mass index, kg/m ²	28.5 ± 5.0	29.0 ± 4.9	28.7 ± 5.0
Current or former smoking	31 (47.0)	35 (53.8)	66 (50.4)
Comorbidities			
Hypertension	41 (62.1)	50 (76.7)	91 (69.5)
Diabetes mellitus	10 (15.2)	16 (24.6)	26 (19.8)
Hypercholesterolemia	54 (81.8)	44 (67.7)	98 (74.8)
Previous myocardial infarction	2 (3.0)	5 (7.7)	7 (5.3)
Previous percutaneous coronary intervention	5 (7.6)	11 (16.9)	16 (12.2)
Canadian Cardiovascular Society angina class			
I	2 (3.0)	1 (1.5)	3 (2.3)
II	43 (65.2)	39 (60)	82 (62.6)
III	21 (31.8)	25 (38.5)	46 (35.1)
Angina duration, mo	6 (3.5-10.5)	6 (3-9)	5 (4-12)
Positive functional test results at enrollment			
Dobutamine stress echocardiography	13 (19.7)	11 (16.9)	24 (18.3)
Magnetic resonance imaging perfusion	0	1 (1.5)	1 (0.8)
Exercise tolerance test	17 (25.8)	12 (18.5)	29 (22.1)

Values are mean ± SD, n (%), or median (Q1-Q3).

translesional pressure assessment ($FFR \leq 0.80$ or $iFR \leq 0.89$). All patients in the PCI arm received drug-eluting stents. Procedural demographics are shown in [Table 2](#). Most lesions were in the left anterior descending coronary artery ($n = 103$ of 131 [78.6%]) and were frequently in the proximal or ostial vessel ($n = 70$ of 131 [53.4%]). The mean FFR was 0.69 ± 0.16 , and 96 patients (96 of 131 [73.3%]) had $FFR \leq 0.80$. The mean FFR in the subgroup of patients with FFR -significant disease ($FFR \leq 0.80$) was 0.63 ± 0.13 ; in the remainder, this value was 0.87 ± 0.04 . For computed virtual physiology, per-vessel and population distributions of MVR_{CFD} and hyperemic inlet and outlet flow are shown in [Supplemental Figures 2 and 3](#). The median MVR_{CFD} was $1.38 \text{ mm Hg} \cdot \text{min/mL}$ (Q1-Q3: 0.89 - $2.09 \text{ mm Hg} \cdot \text{min/mL}$). Patients were followed for a median 6 weeks (Q1-Q3: 6-6 weeks) after randomization.

STUDY ENDPOINTS. Exercise time. At baseline, there was mild evidence of a relationship between MVR_{CFD} and exercise time (ordinal correlation coefficient = 0.20; 95% CrI: 0.18-0.22). Paired pre-randomization and follow-up exercise time data were available for 129 patients (63 PCI, 66 placebo). The mean prerandomization exercise time was 495 ± 188 seconds. For a patient with low MVR_{CFD} ($0.77 \text{ mm Hg} \cdot$

TABLE 2 Procedural Demographics

	Percutaneous Coronary Intervention (n = 66)	Placebo (n = 65)	Complete Group (N = 131)
Lesion characterization			
Left anterior descending coronary artery	53 (80.3)	50 (76.9)	103 (78.6)
First diagonal branch	1 (1.5)	1 (1.5)	2 (1.5)
Left circumflex coronary artery	2 (3.0)	7 (10.8)	9 (6.9)
Obtuse marginal branch	2 (3.0)	0	2 (1.5)
Intermediate branch	1 (1.5)	1 (1.5)	2 (1.5)
Right coronary artery	7 (10.6)	6 (9.2)	13 (9.9)
Ostial/proximal	44 (66.7)	26 (40.0)	70 (53.4)
Mid	20 (30.3)	34 (52.3)	54 (41.2)
Distal	2 (3.0)	5 (7.7)	7 (5.3)
Serial lesions	10 (15.2)	8 (12.3)	18 (13.7)
Area stenosis by quantitative coronary angiography, %	85.3 ± 12.5	85.5 ± 11.2	85.4 ± 11.9
Diameter stenosis by quantitative coronary angiography, %	65.3 ± 16.1	66.1 ± 14.9	65.3 ± 15.5
FFR	0.69 ± 0.15	0.69 ± 0.16	0.69 ± 0.16
Median (Q1-Q3)	0.71 (0.61-0.81)	0.72 (0.57-0.83)	0.71 (0.59-0.81)
iFR	0.86 ± 0.12	0.82 ± 0.17	0.84 ± 0.15
Median (Q1-Q3)	0.89 (0.84-0.93)	0.89 (0.81-0.92)	0.89 (0.83-0.92)
Number of patients with FFR ≤ 0.80	50 (75.8)	46 (70.8)	96 (73.3)
Number of patients with iFR ≤ 0.89	33 (50.0)	33 (50.8)	66 (50.4)
Procedural details			
Stent length, mm	23.0 (16.0-30.0)	—	—
Stent diameter, mm	3.00 (2.75-3.50)	—	—
FFR post-PCI	0.89 ± 0.06	—	—
Median (Q1-Q3)	0.88 (0.86-0.93)	—	—
iFR post-PCI	0.94 ± 0.04	—	—
Median (Q1-Q3)	0.95 (0.92-0.96)	—	—
Number of patients with post-PCI FFR >0.80	61 (92.4)	—	—
Number of patients with post-PCI iFR >0.89	60 (88.9)	—	—

Values are n (%) or mean ± SD unless otherwise indicated.
FFR = fractional flow reserve; iFR = instantaneous wave-free ratio; PCI = percutaneous coronary intervention.

min/mL), PCI increased exercise time by a median of 48 seconds vs placebo (95% CrI: 6-92 seconds; Pr = 98.5%). For a patient with high MVR_{CFD} (2.43 mm Hg · min/mL), PCI increased exercise time by a median of 16 seconds (95% CrI: -29 to 61 seconds; Pr = 75.2%) vs placebo ([Table 3](#)). There was modest evidence for this interaction (Pr_{interaction} = 83.1%) ([Central Illustration](#), [Supplemental Appendix](#)).

DSE. At baseline there was no evidence of a relationship between MVR_{CFD} and DSE score (ordinal correlation coefficient = 0.015; 95% CrI: -0.038 to 0.047). Paired prerandomization and follow-up dobutamine stress echocardiographic data were available for 107 patients (48 PCI, 59 placebo). The mean prerandomization DSE score was 1.55 ± 1.75. For a patient with low MVR_{CFD}, PCI decreased DSE score by a median of 0.83 vs placebo (95% CrI: -1.42 to -0.30; Pr = 99.9%). For a patient with high MVR_{CFD}, PCI decreased DSE score by a median of 0.10 (95% CrI: -0.63 to 0.42; Pr = 65.9%) vs placebo ([Table 3](#)). There was good evidence for this interaction (Pr_{interaction} = 96.5%) ([Central Illustration](#), [Supplemental Appendix](#)).

Patient-reported symptoms and quality of life.

At baseline, there was no evidence of a relationship between MVR_{CFD} and freedom from angina (ordinal correlation coefficient = 0.17; 95% CrI: -0.12 to 0.33). Paired prerandomization and follow-up data from the patient-reported SAQ were available in 127 patients (63 PCI, 64 placebo). At prerandomization, 105 patients (82.7%) had angina. For a patient with low MVR_{CFD}, PCI increased the log odds of complete freedom from angina by a median 1.24 vs placebo (95% CrI: 0.16-2.38; Pr = 98.8%). For a patient with high MVR_{CFD}, PCI increased the log odds of complete freedom from angina by a median of 0.10 (95% CrI: -1.07 to 1.25; Pr = 57.1%) vs placebo ([Table 3](#)). There was modest evidence for this interaction (Pr_{interaction} = 90.0%) ([Figure 1A](#), [Supplemental Appendix](#)).

At baseline, there was weak evidence of a relationship between MVR_{CFD} and SAQ angina frequency (ordinal correlation coefficient = 0.12; 95% CrI: 0.01-0.15). The mean prerandomization SAQ angina frequency score was 71.7 ± 23.4 points. For a patient with low MVR_{CFD}, PCI improved angina frequency score by

TABLE 3 Bayesian Analysis of Primary and Secondary Endpoints

	Low Microvascular Resistance	High Microvascular Resistance
Exercise time (seconds)		
n		129
Baseline mean		495 ± 188
Correlation with MVR _{CFD}		0.203 (95% CrI: 0.184-0.218)
Incremental effect of PCI over placebo	48 (6-92)	16 (–29 to 61)
Probability of significant benefit favoring PCI, %	98.5	75.2
Probability of interaction, %		83.1
Dobutamine stress echocardiography score		
n		107
Baseline mean		1.04 (0.42-2.02)
Correlation with MVR _{CFD}		0.015 (95% CrI: –0.038 to 0.047)
Incremental effect of PCI over placebo	–0.83 (–1.42 to –0.30)	–0.10 (–0.63 to 0.42)
Probability of significant benefit favoring PCI, %	99.9	65.9
Probability of interaction, %		96.5
SAQ freedom from angina		
n		127
Angina present		105 (82.7)
Correlation with MVR _{CFD}		0.165 (95% CrI: –0.117 to 0.327)
Log odds of complete freedom from angina	1.24 (0.16-2.38)	0.10 (–1.07 to 1.25)
Probability of significant benefit favoring PCI, %	98.8	57.1
Difference in log odds ratios		1.14 (95% CrI: –0.60 to 2.90)
Probability of interaction, %	90.0	
SAQ angina frequency score		
n		127
Baseline mean		71.7 ± 23.4
Correlation with MVR _{CFD}		0.120 (95% CrI: 0.097-0.148)
Incremental effect of PCI over placebo	7.67 (0.25-15.6)	–0.25 (–9.24 to 8.80)
Probability of significant benefit favoring PCI, %	97.8	47.8
Probability of interaction, %		89.2
SAQ physical limitation score		
n		114
Baseline mean		67.7 ± 24.7
Correlation with MVR _{CFD}		0.257 (95% CrI: 0.250-0.258)
Incremental effect of PCI over placebo	7.88 (–0.01 to 15.5)	8.96 (–0.92 to 17.8)
Probability of significant benefit favoring PCI, %	97.9	96.6
Probability of interaction, %		43.6
SAQ treatment satisfaction score		
n		127
Baseline mean		86.4 ± 15.4
Correlation with MVR _{CFD}		0.043 (95% CrI: –0.061 to 0.081)
Incremental effect of PCI over placebo	0.92 (–3.84 to 5.59)	–2.46 (–8.44 to 2.91)
Probability of significant benefit favoring PCI, %	65.3	19.0
Probability of interaction, %		80.2
SAQ quality-of-life score		
n		126
Baseline mean		48.4 ± 22.2
Correlation with MVR _{CFD}		0.115 (95% CrI: 0.062-0.147)
Incremental effect of PCI over placebo	2.51 (–7.50 to 12.6)	0.07 (–10.9 to 11.4)
Probability of significant benefit favoring PCI, %	68.7	50.5
Probability of interaction, %		61.2
SAQ stability score		
n		126
Baseline mean		64.5 ± 25.8
Correlation with MVR _{CFD}		0.196 (95% CrI: 0.150-0.213)
Incremental effect of PCI over placebo	0.36 (–10.9 to 11.1)	–4.53 (–16.5 to 7.23)
Probability of significant benefit favoring PCI, %	52.7	22.6
Probability of interaction, %		29.6

Continued on the next page

TABLE 3 Continued

	Low Microvascular Resistance	High Microvascular Resistance
EQ-5D-5L quality-of-life score		
n	127	
Baseline mean	0.79 ± 0.21	
Correlation with MVR _{CFD}	0.210 (95% CrI: 0.184-0.243)	
Incremental effect of PCI over placebo	0.00 (−0.05 to 0.06)	−0.02 (−0.08 to 0.04)
Probability of significant benefit favoring PCI, %	57.0	27.8
Probability of interaction, %	69.7	
EQ-5D-5L visual analogue score		
n	129	
Baseline mean	65.6 ± 20.8	
Correlation with MVR _{CFD}	0.139 (95% CrI: 0.105-0.162)	
Incremental effect of PCI over placebo	0.79 (−6.10 to 7.99)	3.05 (−4.99 to 10.5)
Probability of significant benefit favoring PCI, %	58.8	78.4
Probability of interaction, %	34.7	
CCS angina classification		
n	129	
Baseline mean	1.94 ± 0.93	
Median (Q1-Q3)	2 (2-3)	
Correlation with MVR _{CFD}	0.259 (95% CrI: 0.208-0.266)	
Incremental effect of PCI over placebo	0.10 (−0.41 to 0.64)	−0.05 (−0.57 to 0.48)
Probability of significant benefit favoring PCI, %	35.8	57.0
Probability of interaction, %	63.7	

CCS = Canadian Cardiovascular Society; MVR_{CFD} = microvascular resistance computed from computational fluid dynamics; PCI = percutaneous coronary intervention; SAQ = Seattle Angina Questionnaire.

a median of 7.67 points vs placebo (95% CrI: 0.25-15.6 points; Pr = 97.8%). For a patient with high MVR_{CFD}, placebo improved angina frequency score by a median of 0.25 points (95% CrI: −9.24 to 8.80 points; Pr = 47.8%) vs PCI (Table 3). There was modest evidence for this interaction (Pr_{interaction} = 89.2%) (Figure 1B, Supplemental Appendix).

At baseline, there was mild evidence of a relationship between MVR_{CFD} and SAQ physical limitation score (ordinal correlation coefficient = 0.257; 95% CrI: 0.250-0.258). The mean prerandomization SAQ physical limitation score was 67.7 ± 24.7 points. For a patient with low MVR_{CFD}, PCI improved physical limitation score by a median of 7.88 points vs placebo (95% CrI: −0.01 to 15.5 points; Pr = 97.9%). For a patient with high MVR_{CFD}, PCI improved physical limitation score by a median of 8.96 points (95% CrI: −0.92 to 17.8 points; Pr = 96.6%) vs placebo (Table 3). There was no evidence of an interaction between MVR_{CFD} and improvement in SAQ physical limitation score (Pr_{interaction} = 43.6%) (Supplemental Figure 4).

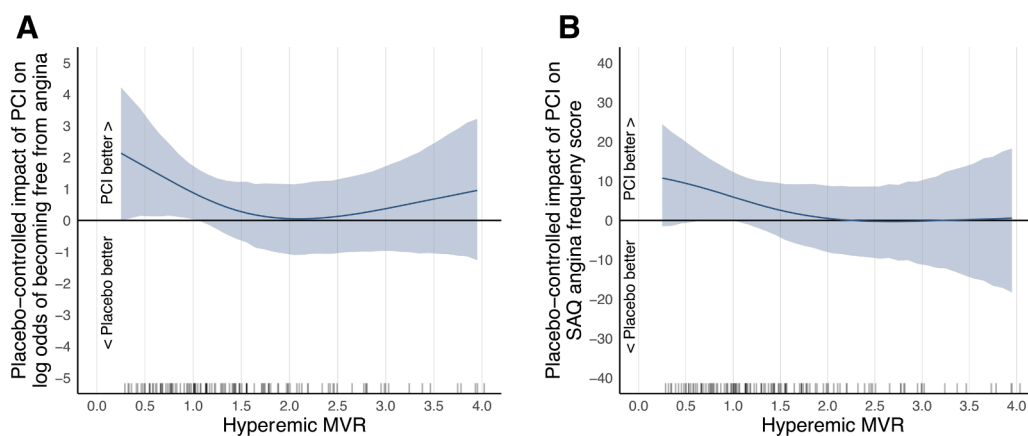
There was no strong evidence of an interaction with MVR_{CFD}, or a difference in treatment outcome in low- vs high-MVR_{CFD} patients, for SAQ treatment satisfaction (Pr_{interaction} = 80.2%), SAQ quality of life (Pr_{interaction} = 61.2%), SAQ angina stability

(Pr_{interaction} = 29.6%), EQ-5D-5L quality of life (Pr_{interaction} = 69.7%), or EQ-5D-5L visual analogue scale score (Pr_{interaction} = 34.7%) (Table 3, Supplemental Figures 5 to 9).

Physician-assessed symptoms. At baseline, there was mild evidence of a relationship between MVR_{CFD} and physician-assessed CCS class (ordinal correlation coefficient = 0.26; 95% CrI: 0.21-0.27). Paired prerandomization and follow-up physician-assessed CCS class data were available for 129 patients (63 PCI, 66 placebo). The median prerandomization CCS class was II (Q1-Q3: II-III). For a patient with low MVR_{CFD}, PCI increased CCS class by a median of 0.10 vs placebo (95% CrI: −0.41 to 0.64; Pr = 35.8%). For a patient with high MVR_{CFD}, PCI decreased CCS class by a median of 0.05 (95% CrI: −0.48 to 0.57; Pr = 57.0%) vs placebo (Table 3). There was no evidence of an interaction between MVR_{CFD} and improvement in physician-assessed CCS class (Pr_{interaction} = 63.7%) (Supplemental Figure 10).

DISCUSSION

This study provides the first placebo-controlled data on the association between CMD and the benefits of PCI. The results suggest a relationship between microvascular physiology and response to PCI that is

FIGURE 1 Relationship of Change in Prerandomization to Follow-Up SAQ Scores Against Prerandomization MVR by Randomization Arm

(A) Relationship of change in prerandomization to follow-up total log odds of complete freedom from angina and prerandomization microvascular resistance (MVR) by randomization arm (probability of interaction = 90.0%). (B) Relationship of change in prerandomization to follow-up total Seattle Angina Questionnaire (SAQ) angina frequency score and prerandomization MVR by randomization arm. At progressively lower MVR values, there was a progressively larger difference between PCI and placebo on the endpoint (probability of interaction = 89.2%). At progressively lower MVR values, there was a progressively larger difference between PCI and placebo on the endpoint.

both biologically cogent and consistent across multiple outcome measures. Low MVR_{CFD} was associated with benefit of PCI vs placebo for exercise time, likelihood of complete freedom from angina, reduction in angina frequency, and DSE score, but the evidence for interactions with symptom-based endpoints was modest. For all these outcome variables, the presented results suggest that patients with high MVR_{CFD} (80th centile, indicating worse CMD) are unlikely to gain benefit from PCI vs placebo.

ABILITY OF MVR_{CFD} TO PREDICT PRIMARY AND SECONDARY OUTCOMES. ORBITA highlighted the need for placebo-controlled randomized trials in the treatment of stable angina. PCI was not associated with any significant improvement in exercise time, beyond the effect of the placebo,⁵ and stratification according to FFR and iFR did not help select patients who might symptomatically benefit from PCI.⁶ The mechanism by which relief of occlusive epicardial stenoses translates to improvement in patient exercise capacity is complex.⁸ Despite randomization and the presence of maximal antianginal medication, it is possible that concomitant CMD may have attenuated the benefit of PCI for some patients in ORBITA.⁵ The results of the present subanalysis suggest that this may be true; in patients with low MVR_{CFD} , PCI offered an improvement in exercise time of about 50 seconds vs placebo, with a 97.1% chance of this representing benefit favoring PCI vs placebo. Although this is the first evidence of an interactional effect on

the placebo-controlled effect of PCI on exercise time in patients treated with antianginal medication, the expected incremental improvement in exercise time was lower than that attributed to balloon angioplasty in the unblinded ACME (Angioplasty Compared to Medicine) trial.⁴ It is therefore likely that the placebo effect of PCI remains significant even in those with no evidence of CMD.

Computed MVR was also associated with several patient-centered secondary endpoints. The strongest association was with complete freedom from angina, which is often the ultimate goal of treatment. Of those included in this subanalysis, a patient with low (20th centile) MVR_{CFD} might expect a 3.5-fold greater likelihood of complete freedom from angina when treated with PCI vs placebo. Contrastingly, these benefits of PCI almost disappear in patients with high MVR_{CFD} . Even if symptoms did not resolve completely, our results suggest that PCI may achieve a significant reduction in angina frequency for patients with low MVR_{CFD} . There was also significant improvement in the SAQ physical limitation score in patients with low MVR_{CFD} , but accounting for patients with high MVR_{CFD} , there was no evidence of an interactional effect.

An association was also observed between MVR_{CFD} and the placebo-controlled improvement in DSE score, with greater improvement in patients with low MVR_{CFD} . The translesional indexes FFR and iFR have previously been shown to predict DSE improvement in response to PCI.^{5,8} However, these new results

suggest a distinct mechanism relating stress-induced myocardial ischemia and regional wall abnormalities with CMD. This may be caused by structural microvascular remodeling, limiting myocardial flow, or chronic repeated ischemic injury uncoupling perfusion from contractility. As there was strong evidence for the interaction with DSE score improvement, it is plausible that CMD may have been confined to a specific coronary territory in the present study. Given that DSE improvements are also associated with epicardial lesion severity,⁶ patients with physiologically significant CAD and healthy microvasculature may expect the largest improvement in DSE score. This aligns well with current understanding of microvascular physiology, but the present data lack power to investigate for a secondary interactional effect. Furthermore, on a background of less intense medical therapy, the epicardial over microvascular contribution to angina provoking ischemia may be more significant, extending the benefits of PCI to patients with greater microvascular dysfunction.

CLINICAL IMPLICATIONS. In recent years, the significance of CMD for myocardial ischemia has been drawn into sharp focus.^{9,18} Current guidelines for the treatment of chronic coronary syndromes place considerably more emphasis on angina with no obstructive CAD than previous versions and recognize distinct CMD phenotypes.^{1,19} However, diagnosis and management remain focused largely on the exclusion of significant CAD.²⁰ The results of this study suggest an important, biologically plausible, relationship between MVR_{CFD} and response to PCI in those with significant epicardial stenoses. Although the clinical relevance of the relatively small improvement in exercise time reported in this study (48 seconds) is uncertain, those with low (20th centile) MVR_{CFD} also derived a 3.5-fold greater likelihood of complete freedom from angina when treated with PCI vs placebo. The ability to better identify those likely to derive benefit from PCI and those who may be better suited to medical therapy alone is clearly beneficial but requires prospective evaluation, including MVR diagnostic threshold determination for dichotomizing patients.

ORBITA included an intensive antianginal medication optimization phase, with an average of 2.8 antianginal agents per patient.^{1,21} Levels of antianginal medications were comparable between both arms of the present subanalysis, but this level of medical therapy may not reflect routine practice in many health care systems. This, combined with the inclusion of patients only with single-vessel disease, may have introduced a ceiling effect for the potential

benefit of PCI. Our results showed a large spread of MVR_{CFD} among included patients and indicate that CMD attenuated the beneficial effect of PCI in several instances. We have shown that patients with low MVR_{CFD} may have gained an increase in treadmill time of approximately 50 seconds with PCI. This incremental improvement is close to the placebo-controlled PCI-mediated 60-second improvement in exercise time in patients taking no antianginal medications.²² Therefore, in patients with low MVR_{CFD}, who may be expected to gain maximum benefit of revascularization, the beneficial effects of PCI on exercise time may only be partially attenuated by antianginal therapy. Patients with moderate or high MVR_{CFD} gained less improvement in exercise time from PCI compared with the average expected benefit of 48 to 55 seconds with antianginal monotherapy.^{23,24}

ACCURACY OF THE CFD TECHNIQUE. The model of MVR_{CFD} has undergone benchtop¹³ and, recently, multicenter in vivo validation against the current invasive gold standard of continuous infusion thermodilution.²⁵ As this invasive comparator cannot quantify absolute resistance in the distal vessel, downstream of epicardial stenoses, the most appropriate microvascular-specific index for CFD validation is MVR reserve. This validation analysis included 116 vessels in patients with epicardial disease and found a moderate to strong relationship with continuous thermodilution for MVR reserve ($\rho = 0.58$; $P < 0.0001$). Furthermore, the area under the curve for diagnosing CMD at a predetermined threshold of 2.5 was 0.77 (95% CI: 0.68-0.86). These data suggest a comparable relationship and level of accuracy as that observed between Doppler with bolus thermodilution ($r = 0.43$; $P < 0.0001$)¹¹ and bolus with continuous infusion thermodilution ($r = 0.26$; $P < 0.001$),²⁶ all of which are used in modern clinical practice. Nevertheless, uncertainties in geometric reconstruction accuracy and the method for implementation of side branch flow remain significant contributors to error in the present model.^{27,28}

STUDY LIMITATIONS. This MVR-stratified analysis of ORBITA included only 2 thirds of trial patients. The statistical analysis plan was designed to maximize power with available data, but further work is required to confirm these findings. The CFD tool does not normalize for myocardial mass. Post-PCI iFR and FFR indicated that some patients may have had residual ischemia, and the possibility that this may have been due to potentially treatable focal lesions cannot be excluded.²⁹ Further research will

demonstrate whether the observed interactions are preserved in those taking fewer antianginal agents. The absence of high MVR_{CFD} may not necessarily mean that a patient's chest pain was caused by epicardial disease. Evidence for guiding optimal management of CMD is relatively sparse¹ and is the subject of ongoing research (NCT04606459, NCT05102019).

CONCLUSIONS

In this MVR-stratified analysis of ORBITA, progressively lower MVR_{CFD} was associated with a larger benefit of PCI vs placebo for exercise time, likelihood of complete freedom from angina, frequency of anginal symptoms, and improvement in DSE score. For patients with single-vessel stenoses already taking optimal antianginal medications, these results suggest that CMD may attenuate the response to PCI in terms of coronary blood flow, exercise capacity, and patient-centered outcomes.

DATA AVAILABILITY All generated data are included in the paper and supplemental material.

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PERSPECTIVES

WHAT IS KNOWN? CMD is an increasingly recognized cause of myocardial ischemia.

WHAT IS NEW? In the presence of functionally significant epicardial stenoses, computed MVR was associated with the symptomatic and functional response to PCI in ORBITA.

WHAT IS NEXT? The interaction between computed or measured MVR and PCI should be further evaluated, perhaps in the context of revascularization given as monotherapy or alongside less aggressive medical therapy regimens.

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APPENDIX For supplemental tables, methods, results, and figures, please see the online version of this paper.